1. Introduction to Animal Feeding and Commercial Animals

- **Commercial Animals:** Animals kept and reared by humans for economic benefits. Examples include cattle, goats, fowl, pigs, turkeys, rabbits, sheep, and fish.
- **Animal Feed:** Food items prepared by the farmer and given to farm animals for growth, reproduction, energy, and overall health.
- **Animal Ration:** The total amount of feed given to an animal within a 24-hour period to meet its specific nutritional needs.

2. Classification of Animal Feed

Animal feeds are broadly classified into four major categories:

- **Basal Feed (Energy Feed):** * Starchy foods high in carbohydrates but low in protein.
  - Used for both ruminants and non-ruminants (monogastrics) like pigs and poultry.
  - Sources include maize, rice, millet, guinea corn, wheat, cassava, yam, and sweet potato.
- **Concentrates:** * Highly digestible mixtures that are rich in proteins, carbohydrates, minerals, and vitamins, but low in fiber.
  - Fed in small amounts mainly to non-ruminant animals like pigs and poultry.
  - Ingredients include cereals, fish meal, bone meal, oyster shell meal, soya bean meal, and vitamin premixes.
- **Roughages:** * Feeds containing high amounts of fiber, bulky in nature, and not easily digested by non-ruminants.
  - They supply energy, vitamins, and minerals.
  - Chiefly fed to ruminants (cattle, sheep, goats) and some herbivorous non-ruminants (rabbits, grasscutters).
  - Forms include **forage crops** (green pasture, grasses, legumes), **dry roughages** (straw, hay), **silage** (partially dried green fodder stored in silos), and **brans** (maize, rice, or wheat bran).
- **Supplements:** * Feedstuffs used to supply essential nutrients (like vitamins and minerals) that may be lacking in the primary diet.
  - They promote growth and health, and can be given to all categories of livestock.
  - Examples include salt licks, bone meals, oyster shell meals, and cotton seed or groundnut cakes.

3. Major Nutrients in Animal Feeds

Animal feeds must contain six primary nutrient components:

Nutrient	Main Functions	Common Sources
Carbohydrates	Provides energy; can be converted to fat to fatten the animal.	Cereals, root/tuber crops, forage crops, brans.
Protein	Body growth; tissue repair; production of meat, milk, and eggs; synthesis of sperm and ovum.	Legumes, fish meal, cotton seed cake, copra cake, palm kernel cake, insects.
Fats and Oils	Provides energy (twice the amount of carbohydrates); maintains body temperature; protects internal organs; improves feed taste.	Palm oil, groundnut cake, cotton seed cake, coconut cake, palm kernel cake.
Minerals	Disease prevention; formation of teeth, bone, and blood; regulates body fluid, pH, and osmotic pressure; acts as catalysts.	Salt lick, bone meal, oyster shell meal, forage crops.
Vitamins	Promotes growth; proper metabolism; supports the immune system; aids in reproduction/fetal development; egg and milk production.	Fish meal, grasses, vegetables, yellow maize, green plants.
Water	Essential for survival, metabolic function, temperature regulation, lubrication, and excretion.	Rivers, streams, wells, dams, rainwater, pipe-borne water, and moisture in food.

4. Usefulness of Feed and Water

Importance of Feed

1. **Provides Nutrients:** Nutrient breakdown allows absorption into the blood to keep animals healthy and capable of performing life functions.
2. **Promotes Growth:** Balanced food increases body size and weight quickly.
3. **Generates Energy:** Supplies energy required for movement, physical labor (e.g., ploughing), and normal bodily functions.
4. **Promotes Good Health:** Proper feeding enables the body to produce antibodies to fight off diseases.
5. **Increases Productivity & Profit:** Quality feed ensures the production of quality goods (meat, eggs, milk, wool, leather), maximizing a farmer's profit.
6. **Aids Reproduction:** Nutrient elements help produce healthy reproductive cells (sperms and ovum) to yield healthy young ones.
7. **Maintains Body Structure:** Helps retain the ideal body shape and structure of the farm animal.

Importance of Water

1. **Major Body Constituent:** Water forms more than half of an animal's body and is a major component of blood, milk, eggs, and flesh.
2. **Facilitates Metabolic Processes:** Lubricates food traveling through the digestive tract and aids in digestion, chewing, swallowing, and nutrient distribution.
3. **Prevents Dehydration:** Maintains normal functioning of cells, tissues, and organs.
4. **Excretion of Waste:** Acts as a solvent to dissolve and transport metabolic waste products out of the body via feces, sweat, and urine.
5. **Regulates Body Temperature:** Facilitates heat loss through evaporation (skin and lungs) and evenly redistributes heat throughout the body via blood circulation.
6. **Promotes Reproduction:** Required for the development and formation of sperm and ovum.
7. **Lubricant:** Keeps parts of the body (like the eyes) moist, lowering the risk of injury.

Effects of Water Inadequacy

- Animals suffer from **dehydration**, causing their excretion and metabolism systems to stall, leading to tissue and organ malfunction.
- Animals look sick, weak, emaciated, and lose market appeal.
- Production of milk, eggs, or meat decreases significantly, and long periods of dehydration ultimately cause **death**.

5. Proportional Feed Requirements

- Nutritional requirements vary according to an animal's **age, sex, breed, growth stage,** and **physical condition** (e.g., whether they are sick, pregnant, lactating, or used for heavy farm labor).
- **Pregnant/Lactating Animals:** Require elevated proportions of protein, iron, and calcium to keep themselves healthy and protect their young ones.
- **Young Males:** Require extra energy, protein, vitamins, and minerals for growth and reproductive activities.
- **Broilers vs. Layers:** Broilers (raised for meat) and Layers (raised for egg production) have different diets. Layers require considerably **more proteins, fats and oils, calcium, and minerals** in their ration compared to broilers to ensure the production of healthy eggs with robust shells.